CURRICULUM VITAE

- **Personal Details**

Ohad Elishco

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Ben-Gurion University of the Negev
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ORCID iD: 0000-0002-8551-1592

- **Education**

B.Sc. 2008 – 2012
Ben-Gurion University of the Negev, Israel
Department of Electrical and Computer Engineering

B.Sc. 2008 – 2012
Ben-Gurion University of the Negev, Israel
Department of Mathematics

M.Sc. Magna Cum Laude 2011 – 2013
Ben-Gurion University of the Negev, Israel
Department of Electrical and Computer Engineering
Advisor: Prof. Haim H. Permuter
Thesis: "Capacity And Coding For The Ising Channel With Feedback"

Ph.D 2013 – 2017
Ben-Gurion University of the Negev, Israel
Department of Electrical and Computer Engineering
Advisors: Prof. Moshe Schwartz and Prof. Tom Meyerovitch
Thesis: "Semiconstrained Systems"

- **Employment History**

Senior Lecturer 2020 – Present
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

Post-Doctoral Researcher 2018 – 2020
Department of Electrical and Computer Engineering
University of Maryland, College Park, USA
Host: Prof. Alexander Barg

Post-Doctoral Researcher 2017 – 2018
Research Laboratory of Electronics
Department of Electrical and Computer Engineering
Massachusetts Institute of Technology
Host: Prof. Muriel Médard

- **Professional Activities**

(a) Positions in academic administration

Head of the mathematical core track — 2022 – Present
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

Head of Math-EE Dual degree Track — 2022 – Present
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

Member of Departmental Seminar committee — 2022 – 2024
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

Departmental education liaison person to math courses — 2024 – present
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

(b) Professional functions outside universities/institutions

Technical-Program Committee Member

IEEE Information Theory Workshop[1] (ITW2025)
IEEE International Symposium on Information Theory[1] (ISIT2025)
IEEE International Conference on Communications (ICC 2025)
IEEE International Symposium on Information Theory[1] (ISIT2024)
IEEE International Conference on Communications (ICC 2024)
IEEE Global Communications Conference (GLOBECOM2023)
IEEE International Conference on Communication (ICC 2023)
IEEE Global Communications Conference (GLOBECOM2022)
IEEE International Symposium on Information Theory[1] (ISIT2022)
IEEE Global Communications Conference (GLOBECOM2021)

Reviewer

ISF – Israel Science Foundation

(c) Significant professional consulting

(d) Editor or member of editorial board of scientific or professional journal

(e) Ad-hoc reviewer for journals and conferences

IEEE Transactions on Communications
IEEE Transactions on Information Theory

[1] TPC position involves more than paper reviews; it includes assigning reviewers and conducting a meta-review.

IEEE International Symposium on Information Theory
IEEE Information Theory Workshop
International Society for Computational Biology
Mathematical Reviews
IEEE/ACM Transactions on Computational Biology and Bio-informatics
Discrete Mathematics
Journal of Combinatorial Theory
IEEE Transactions on Molecular, Biological, and Multi-Scale Communications
Designs, Codes and Cryptography

(f) Membership in professional/scientific societies

Senior Member — 2024 – Present
IEEE Information Theory Society

Member — 2017 – 2024
IEEE Information Theory Society

Student Member — 2012 – 2017
IEEE Information Theory Society

- **Educational activities**

(a) Courses taught

Laboratory in Computer Architecture
Ben-Gurion University of the Negev, Israel
Undergraduate compulsory Lab
(Lab Instructor)

Introduction to Computers
Ben-Gurion University of the Negev, Israel
Undergraduate compulsory course
(TA)

Modern Discrete Probability
University of Maryland, College Park, USA
Graduate elective course

Digital Systems
Ben-Gurion University of the Negev, Israel
Undergraduate compulsory course

Introduction to Coding for Constrained Systems
Ben-Gurion University of the Negev, Israel
Graduate elective course

Modern Discrete Probability for Information Theory, Big Data, and Machine Learning
Ben-Gurion University of the Negev, Israel
Graduate elective course

Introduction to control theory

Ben-Gurion University of the Negev, Israel
Undergraduate elective course

(b) Research students

M.Sc. Students

Yair Yaakobi — Graduated 2024
Department of Electrical and Computer Engineering
Tel-Aviv University, Israel
(Jointly supervised with Prof. Itzhak Tamo, EE, TAU)

Nimrod Arcusin — Graduated 2025
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

Noam Klainer — Expected 2026
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

Liav Koram — Expected 2026
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

Hagai Berend — Expected 2026
School of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel
(Jointly supervised with Prof. Moshe Schwartz, EE, BGU)

Bessart Dalloma — Graduated 2025
Department of Computer Science
Technion - Israel Institute of Technology, Israel
(Jointly supervised with Prof. Eitan Yaakobi, CS, Technion)

Ph.D. Students

Post-Doctoral Students

Zuo Ye — 2022–2025
Department of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

- **Awards, Citations, Honors, Fellowships**

(a) Honors, Citation Awards

Head of ECE Department's honor — 2010
Ben-Gurion University of the Negev, Israel

MEITAR direct M.Sc program for Excelling students — 2011
Ben-Gurion University of the Negev, Israel

M.Sc. graduation magna cum laude — 2014
Ben-Gurion University of the Negev, Israel

Negev Scholarship for outstanding Ph.D. students — 2014
Ben-Gurion University of the Negev, Israel

Glatt Award for Excellence in Teaching — 2022
Ben-Gurion University of the Negev, Israel

Dean Award for Excellence in Teaching — 2024
Ben-Gurion University of the Negev, Israel

School Award for Excellence in Teaching — 2025
Department of Electrical and Computer Engineering
Ben-Gurion University of the Negev, Israel

(b) Fellowships

- **Scientific Publications**

H-index: WOS (previously ISI) 6, GS 8.
Total number of citations of all articles: WOS (previously ISI) 167, GS 258.
Total number of citations without self citations: WOS (previously ISI) 148.

(a) Refereed chapters in collective volumes, Conference proceedings, Festschrifts, etc.

1. **Ohad Elishco**[S] and Haim Permuter[PI], "Capacity of the Ising Channel With Feedback," *Proc. of the 2011 IEEE International Symposium on Information Theory, ISIT2011,* pp. 3004–3008, St. Petersburg, Russia, August 2011 ($\sim$ 60% acceptance).
2. **Ohad Elishco**[S], Tom Meyerovitch[PI], and Moshe Schwartz[PI], "Semiconstrained systems," *Proc. of the 2015 IEEE International Symposium on Information Theory, ISIT2015,* pp. 246–250, Hong Kong, China SAR, June 2015 ($\sim$ 60% acceptance).
3. **Ohad Elishco**[S], Farzad Farnoud (Hassanzadeh)[PD], Moshe Schwartz[PI], and Jehoshua Bruck[PI], "The capacity of some Pólya string models," *Proc. of the 2016 IEEE International Symposium on Information Theory, ISIT2016,* pp. 270–274, Barcelona, Spain, July 2016 ($\sim$ 60% acceptance).
4. **Ohad Elishco**[S], Tom Meyerovitch[PI], and Moshe Schwartz[PI], "Encoding semiconstrained systems," *Proc. of the 2016 IEEE International Symposium on Information Theory, ISIT2016,* pp. 395–399, Barcelona, Spain, July 2016 ($\sim$ 60% acceptance).
5. **Ohad Elishco**[S], Tom Meyerovitch[PI], and Moshe Schwartz[PI], "Multidimensional semiconstrained systems," *Proc. of the 2017 IEEE International Symposium on Information Theory, ISIT2017,* pp. 1490–1494, Aachen, Germany, June 2017 ($\sim$ 60% acceptance).
6. W. Huleihel[PD], **Ohad Elishco**[PD], and Muriel Médard[PI], "Blind Group-Testing," *Proc. of the 2018 IEEE International Symposium on Information Theory, ISIT2018,* pp. 2594–2598, Vail, Colorado, USA, June 2018 ($\sim$ 60% acceptance).
7. **Ohad Elishco**[PD], Ryan Gabrys[PD], and Eitan Yaakobi[PI], "Bounds and Constructions of Codes Over Symbol Pair-Read Channels," *Proc. of the 2018 IEEE International Symposium on Information Theory, ISIT2018,* pp. 2505–2509, Vail, Colorado, USA, June 2018 ($\sim$ 60% acceptance).
8. Derya Malak[PD], **Ohad Elishco**[PD], Muriel Médard[PI], and Edmond Yeh[PI], "Throughput and Delay Analysis For Coded ARQ," *Proc. of the International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt-2019),* Avignon, France, June 2019 ($\sim$ 41% acceptance).
9. **Ohad Elishco**[PD], Ryan Gabrys[PD], Eitan Yaakobi[PI], and Muriel médard[PI]"Repeat-Free Codes," *Proc. of the 2019 IEEE International Symposium on Information Theory, ISIT2019,* pp. 932–936, Paris, France, July 2019 ($\sim$ 60% acceptance).

10. **Ohad Elishco**[PD], and Alexander Barg[PI]"Capacity of Dynamical Storage Systems," *Proc. of the 2019 IEEE International Symposium on Information Theory, ISIT2019*, pp. 1562–1566, Paris, France, July 2019 ($\sim$ 60% acceptance).
11. * **Ohad Elishco**[PI], and Alexander Barg[PI]"Capacity and Construction of Recoverable Systems," *Proc. of the 2021 IEEE International Symposium on Information Theory, ISIT2021*, pp. 3273-3278, Melbourne, Australia (Via Zoom), July 2021 ($\sim$ 60% acceptance).
12. * Alexander Barg[PI]and **Ohad Elishco**[PI]and Ryan Gabrys[PI]and Eitan Yaakobi[PI], "Recoverable Systems on Lines and Grids," *Accepted to the 2022 IEEE International Symposium on Information Theory, ISIT2022.* ($\sim$ 60% acceptance)
13. * Zuo Ye[PD], and **Ohad Elishco**[PI], "Codes for Absorption Channel," *Accepted to the 2023 IEEE International Symposium on Information Theory, ISIT2023.* ($\sim$ 60% acceptance)
14. * **Ohad Elishco**[PI], "On the Long-Term Behavior of k-tuples Frequencies in Mutation Systems," *Accepted to the 2024 IEEE International Symposium on Information Theory, ISIT2024.* ($\sim$ 60% acceptance)
15. * Besart Dollma[S]and **Ohad Elishco**[PI], and Eitan Yaakobi[PI], "Coding for Ordered Composite DNA Sequences," *Accepted to the 2025 IEEE International Symposium on Information Theory, ISIT2025.* ($\sim$ 60% acceptance)

(b) Refereed articles and refereed letters in scientific journals

1. **Ohad Elishco**[S], and Haim Permuter[PI], "Capacity and Coding for the Ising Channel With Feedback," *IEEE Trans. on Inform. Theory,* 60(9), pp. 5138–5149, September 2014. (ISI 42/GS 61 citations; IF 0.958; 95/257; Q2)
2. **Ohad Elishco**[S], Tom Meyerovitch[PI], and Moshe Schwartz[PI], "Semiconstrained systems," *IEEE Trans. on Inform. Theory,* 62(4), pp. 1688–1702, April 2016. (ISI 14/GS 19 citations; IF 2.679; 74/262; Q2)
3. **Ohad Elishco**[S], Tom Meyerovitch[PI], and Moshe Schwartz[PI], "On encoding semiconstrained systems," *IEEE Trans. on Inform. Theory,* 64(4), pp. 2474–2484, April 2018. (ISI 4/GS 10 citations; IF 3.215; 38/155; Q1)
4. **Ohad Elishco**[S], Tom Meyerovitch[PI], and Moshe Schwartz[PI], "On independence and capacity of multidimensional semiconstrained systems," *IEEE Trans. on Inform. Theory,* 64(10), pp. 6461–6483, October 2018. (ISI 2/GS 4 citations; IF 3.215; 38/155; Q1)
5. Wasim Huleihel[PD], **Ohad Elishco**[PD], and Muriel Médard[PI], "Blind Group Testing," *IEEE Trans. on Inform. Theory,* 65(8), pp. 5050–5063, August 2019. (ISI 1/GS 1 citations; IF 3.215; 38/155; Q1)
6. **Ohad Elishco**[PD], Farzad Farnoud (Hassanzadeh)[PI], Moshe Schwartz[PI], and Jehoshua Bruck[PI], "The entropy rate of some Pólya string models," *IEEE Trans. on Inform. Theory,* 65(12), pp. 8180–8193, December 2019. (ISI 5/GS 7 citations; IF 3.215; 38/155; Q1)
7. **Ohad Elishco**[PD], Ryan Gabrys[PD], and Eitan Yaakobi[PI], "Bounds and Constructions of Codes Over Symbol Pair-Read Channels," *IEEE Trans. on Inform. Theory,* 66(3), pp. 1385–1395, March 2020. (ISI 26/GS 40 citations; IF 3.215; 38/155; Q1)
8. * **Ohad Elishco**[PI]and Alexander Barg[PI], "Capacity of dynamical storage systems," *IEEE Trans. on Inform. Theory,* 67(1), pp. 329–346, Jan. 2021. (ISI 3/GS 3 citations; IF 2.501; 18/235; Q1)
9. * **Ohad Elishco**[PI]and Ryan Gabrys[PI]and Eitan Yaakobi[PI]and Muriel Médard[PI], "Repeat-Free Codes," *IEEE Trans. on Inform. Theory,* 67(9), pp. 5749–5764, Sept. 2021. (ISI 20/GS 48 citations; IF 2.501; 18/235; Q1)

10. * **Ohad Elishco**[PI] and Alexander Barg[PI], "Recoverable Systems," *IEEE Trans. on Inform. Theory,* 68(6), pp. 3681–3699, June 2022. (ISI 1/GS 9 citations; IF 2.501; 18/235; Q1)
11. * **Ohad Elishco**[PI] and Wasim Huleihel[PI], "Optimal reference for DNA synthesis," *IEEE Trans. on Inform. Theory,* 69(11), pp. 6941–6955, Nov. 2023. (ISI 9/GS 15 citations; IF 2.501; 18/235; Q2)
12. * Zuo Ye[PD] and **Ohad Elishco**[PI], "Reconstruction of a Single String from a Part of its Composition Multiset," *IEEE Trans. on Inform. Theory,* 70(6), pp. 3922–3940, June 2024. (ISI 6/GS 8 citations; IF 2.9; 16/250; Q1)
13. * Zuo Ye[PD] and Wenjun Yu[PD] and **Ohad Elishco**[PI], "Codes over the Absorption Channel," *IEEE Trans. on Inform. Theory,* 70(6), pp. 3981–4001, June 2024. (ISI 2/GS 4 citations; IF 2.9; 16/250; Q1)
14. * Alexander Barg[PI] and **Ohad Elishco**[PI] and Ryan Gabrys[PI] and Eitan Yaakobi[PI] and Geyang Wang[PD], "Storage codes and recoverable systems on lines and grids," *Designs, Codes and Cryptography,* Aug 2024. 92(12), pp. 4145–4168, Dec 2024. (ISI 0/GS 0 citations; IF 1.4; 102/332; Q2)
15. * Xiangliang Kong[PD] and **Ohad Elishco**[PI], "Bounds and Constructions for Generalized Batch Codes," *IEEE Trans. on Inform. Theory,* 70(10), pp. 6857–6876, Oct 2024. (ISI 1/GS 1 citations; IF 2.9; 16/250; Q1)
16. * **Ohad Elishco**[PI], "On the Long-Term Behavior of k-tuples Frequencies in Mutation Systems," *IEEE Trans. on Inform. Theory,* 70(12), pp. 8524–8545, Dec 2024. (ISI 0/GS 2 citations; IF 2.9; 16/250; Q1)
17. * Zuo Ye[PD] and Omer Sabary[S] and Ryan Gabrys[PI] and Eitan Yaakobi[PI] and **Ohad Elishco**[PI], "More on Codes for Combinatorial Composite DNA," *Designs, Codes and Cryptography,* 93(8), pp. 3437-3462, Aug 2025. (ISI 1/GS 1 citations; IF 1.2; 11/89; Q2)
18. * Zuo Ye[PD] and Yubo Sun[S] and Wenjun Yu[PD] and Gennian Ge[PI] and **Ohad Elishco**[PI], "Codes Correcting Two Bursts of Exactly b Deletions," *Accepted to IEEE Trans. on Inform. Theory,* Sep 2025. (ISI 0/GS 5 citations; IF 2.9; 16/250; Q1)

(c) Unrefereed professional articles and publications

- **Lectures and Presentations at Meetings and Invited Seminars**

(a) Invited plenary lectures at conferences/meetings

(b) Presentation of papers at conferences/meetings (oral or poster)[2]

1. **Ohad Elishco**, Tom Meyerovitch, and Moshe Schwartz, "Semiconstrained systems," *The 2015 Information Theory and Applications Workshop, ITA 2015,* San Diego, CA, USA, February 2015. **(invited talk)**
2. **Ohad Elishco**, Tom Meyerovitch, and Moshe Schwartz, "Constrained Error Correcting Coding Schemes," *Israel Mathematical Union Annual Meeting,* Dead-sea, Israel, May 2015. **(invited talk)**
3. **Ohad Elishco**, Tom Meyerovitch, and Moshe Schwartz, "Multidimensional Semiconstrained systems," *The 2017 Information Theory and Applications Workshop, ITA 2017,* San Diego, CA, USA, February 2017. **(invited talk)**

[2]Presentations without proceedings. For presentations with proceedings, see previous section.

4. **Ohad Elishco** and Alexander Barg, "Capacity of Recoverable Systems," *The 2020 Information Theory Workshop, ITW 2020,* Virtual conference, April 2021. **(invited talk)**
5. **Ohad Elishco**, "On Absorption Errors," *Synthetic DNA: From Crispr to Information Storage workshop,* Zichron Ya'akov, Israel, November 2023 (postponed due to war). **(invited talk)**
6. **Ohad Elishco**, "Mutation Systems," *Coding Theory and Algorithms for Emerging Technologies in Synthetic Biology,* Dagstuhl Seminar, Wadern, Germany, December 2024. **(invited talk)**

(c) Presentations at informal international seminars and workshops

(d) Seminar presentations at universities and institutions

1. "Capacity of the Ising Channel with Feedback," Department of Mathematics, Ben-Gurion University, Beer-Sheva, Israel, 2011.
2. "Capacity and Coding for the Ising Channel with Feedback," Department of Electrical and Computer Engineering, Ben-Gurion University, Beer-Sheva, Israel, 2012.
3. "Encoding for Semiconstrained Systems," Department of Computer Science, Technion, Haifa, Israel, 2016.
4. "Semiconstrained Systems," Department of Electrical and Computer Engineering, Ben-Gurion University, Beer-Sheva, Israel, 2017.
5. "Multidimensional Semiconstrained Systems," Department of Electrical and Computer Engineering, MIT, Cambridge, MA, USA, 2017.
6. "Blind Group Testing," Department of Electrical and Computer Engineering, MIT, Cambridge, MA, USA, 2018.
7. "Repeat-Free Codes," Department of Electrical and Computer Engineering, University of Maryland, College Park, MD, USA, 2018.
8. "Coding for Emerging Technologies," School of Electrical and Computer Engineering, Ben-Gurion University, Beer-Sheva, Israel, 2019.
9. "Coding for Emerging Technologies," Department of Electrical and Computer Engineering, Hebrew University, Jerusalem, Israel, 2020.
10. "Coding for Emerging Technologies," Department of Computer Science, Technion, Heifa, Israel, 2020.
11. "Coding for Emerging Technologies," School of Electrical and Computer Engineering, Tel-Aviv University, Tel-Aviv, Israel, 2020.
12. "Coding for Emerging Technologies," Department of Electrical and Computer Engineering, Bar-Ilan University, Ramat-Gan, Israel, 2020.
13. "On Recoverable Systems," Department of Electrical and Computer Engineering, Bar-Ilan University, Ramat-Gan, Israel, 2022.
14. "Reconstruction from composition multiset," Department of Electrical and Computer Engineering, Ben-Gurion University of the Negev, Beer-Sheva, Israel, 2022 (work presented by post-doc).
15. "Codes for Absorption Channel," Department of Computer Science, Technion - Israel technological institute, Haifa, Israel, 2023.

- **Patents**

- **Research Grants**

National Science Foundation and 2021 – 2024
The Israel-United States Binational Science Foundation (NSF-BSF)
US Grantees: Alexander Barg (PI),
Israeli Grantees: Ohad Elishco (PI)
Subject: From storage codes to recoverable systems
Period: 3 Years
US Annual Amount: $183,333
US Total Amount: $500,000
Israeli Annual Amount: $61,500
Israeli Total Amount: $185,071

The Israel Science Foundation (ISF) 2024 – 2028
Grantees: Ohad Elishco (PI)
Subject: Problems in string reconstruction
Period: 4 Years
Annual Amount: $61,000
Total Amount: $244,000
Equipment grant: $ 16,700

- **Present Academic Activities**

Research in progress

1. Random Access Codes.
2. Interspersed mutation systems.
3. graph codes.
4. Average Recoverable Systems.
5. Analysis of Optimal Administration of Antibiotics using mutation systems

Books and articles to be published

In preparation

1. Liav Korem[S] and **Ohad Elishco**[PI], "CLT for mutation systems"
2. **Ohad Elishco**[PI] and Alexander Barg[PI], "The entropy of recoverable and weakly recoverable systems."
3. Noam Klainer[S] and **Ohad Elishco**[PI], "Average synthesis time of DNA sequences."
4. Daniella Bar-Lev[PD] and **Ohad Elishco**[PI] and Ryan Gabrys[PI], "Labeled coupon collector problem."
5. Xiangliang Kong[PD] and **Ohad Elishco**[PI], "On the coupon collector problem with noise."

6. * Avital Boruchovsky[S] and **Ohad Elishco**[PI] and Ryan Gabrys[PI] and Anina Gruica[PD] and Itzhak Tamo[PI] and Eitan Yaakobi[PI], "Optimal distribution for random access codes"

7. * Rami Katz[PI] and Giulia Giordano[PI] and **Ohad Elishco**[PI], "Estimating Colony dynamics using urn models"

8. **Ohad Elishco**[PI], "Interspersed Mutation Systems."

Submitted for publication

1. * Nimrod Arcusin[S] and **Ohad Elishco**[PI], "Repeat-Free Codes with Local Constraints," *submitted to IEEE Trans. on Inform. Theory,* Jul 2025.

2. * Avital Boruchovsky[S] and **Ohad Elishco**[PI] and Ryan Gabrys[PI] and Anina Gruica[PD] and Itzhak Tamo[PI] and Eitan Yaakobi[PI], "Making it to First: The Random Access Problem in DNA Storage," *submitted to IEEE Trans. on Inform. Theory,* Sep 2025.

3. *Besart Dollma[S] and **Ohad Elishco**[PI] and Eitan Yaakobi[PI], "Coding for Ordered Composite DNA Sequences," *submitted to IEEE Trans. on Inform. Theory,* Oct 2025.

4. *Hagai Berend[S] and **Ohad Elishco**[PI] and Moshe Schwartz[PI], "On Multidimensional 2-Weight-Limited Burst-Correcting Codes," *submitted to Designs, Codes, and Cryptography,* Nov 2025.

- **Synopsis of research** [3]

  The volume of data generated daily is experiencing rapid growth and is projected to reach 175 zettabytes by 2025. Accommodating such vast data necessitates unconventional approaches. With data accessibility becoming ubiquitous, information previously stored on standalone systems, such as single hard drives, has migrated to complex systems like cloud storage. These systems consist of numerous simple units undergoing random changes over time, encompassing temporary unit failures or data errors. My primary objectives revolve around comprehending, analyzing, and enhancing the performance of storage systems, with a focus on time-evolving systems. My work is rooted in information/coding theory, with strong connections to symbolic dynamics, ergodic theory, and applied probability. Specifically, my research spans diverse topics, including DNA-based storage systems, distributed storage, and coding for emerging memory devices.

  My scientific journey commenced with an exploration of the theoretical limits of storage systems featuring injected (artificial) uncertainty. Having dual Ph.D. advisors—one from the electrical engineering department and the other from the math department—afforded me the privilege of assimilating two distinct viewpoints (algebraic/combinatoric and dynamical/probabilistic) and acquiring two sets of tools. I am particularly drawn to subjects that amalgamate coding for storage systems and dynamical systems. Accordingly, my research delves into the examination of complex storage systems evolving over time. Employing intuition from each perspective, I amalgamate these toolsets to address challenges. My plan is to deepen the theoretical understanding of such systems while simultaneously progressing toward practical implementation.

  In the subsequent sections, I provide a summary of some contributions made by my collaborators and me in these fields. Additionally, I outline my plans for future research.

  Current and Past Research

  DNA-Based Storage: One prominent solution to the data storage problem is the utilization of DNA-based storage systems as a compact alternative to current digital storage units. Nevertheless, it is not considered a completely reliable storage system, as the information stored in DNA is susceptible to errors. There are two common lines of work on DNA storage systems.

  The first assumes that the information is stored in vivo, i.e., within the DNA of some organisms, and later read from them or their descendants. However, it is corrupted by naturally occurring mutations mathematically modeled by string-duplication systems. In [J6], we delve into the study of the entropy of duplication systems and generalize the process to include noisy duplication. We provide exact expressions for the entropy of end-duplication systems and interspersed-duplication systems, for all noise parameters. This line of inquiry was significantly extended in our recent work [J16]. In this paper, we generalized the duplication system to arbitrary mutation systems and successfully determined the limit (in probability) of k-tuple frequency. This analysis enables us to understand the evolution of such mutation systems and to bound the entropy of the system using an appropriate semi-constrained system (see [J2]).

  The second line of work assumes that information is stored in vitro, i.e., outside of a living organism. Here, errors may occur during the reading process. One method that guarantees a unique reconstruction is to encode the information into a sequence that does not contain any subsequence of length k (for some k) more than once (a k-repeat free sequence). In [C9] and [J9], we provided a full characterization of the storage capacity of those sequences and offered an efficient encoding/decoding algorithm. This study has since evolved into a new, highly practical direction. Recognizing that errors in modern synthesis and sequencing (e.g., nanopore) are

[3] [Jx] (respectively, [Cx]) refer to journal (respectively, conference) publication x.

sequence-dependent, we have developed novel constrained codes that mitigate these errors by design. Our recent (submitted) work [”Repeat-Free Codes with Local Constraints”] provides encoding and decoding algorithms for k-repeat free sequence that also obeys local constraints. This is complemented by our work on [J17] and by the (submitted) work [”oding for Ordered Composite DNA Sequences”], which address other complex error models in composite DNA strands.

Other work in this area includes [J11], where an optimal reference for DNA synthesis was obtained, and [J5], where we estimated the minimum number of tests required to identify unique markers in noisy group testing scenarios.

Distributed Storage Systems: In tandem with the exponential growth of available information, contemporary technologies increasingly gravitate towards online applications that hinge on immediate information access. This necessitates the reliable storage of vast amounts of data in data centers. To achieve this, information is encoded and stored across several units, which may not be inherently reliable.

A parallel challenge in data centers is efficient data retrieval. In our work [J15], we addressed this by designing codes that allow for the efficient retrieval of multiple data items (a ”batch”) simultaneously, thereby minimizing storage redundancy and access latency.

Our work also defined a dynamic model for networks with random failures [J8]. Building upon this, we initiated the study of recoverable systems [J10], which was extended and further explored in [C12, J14], with a specific focus on the analysis of 2D networks. Our ultimate goal is to enable the rapid recovery of lost information with minimal bandwidth usage, leveraging the structural characteristics inherent in the storage network.

Coding for Emerging Memory Devices:

In storage systems, a significant challenge arises from errors caused by inter-symbol interference (ISI). We embarked on the exploration of semiconstrained systems (SCS) [J2, J3, J4], a middle ground that relaxes constraints, allowing limited appearances of error-prone substrings. This allows a less stringent error-correcting code to be applied, reducing the rate penalty. We also addressed ISI through the symbol-pair read channel and the Ising channel models in our works [J1, J7].

Furthermore, our research has pushed into novel communication paradigms. In [J12], we established rigorous bounds and methods for reconstructing data from partial statistical information, relevant for storing data on large molecules like polymers.

A key recent achievement has been in coding for chemical communication. Our work in this area, which began with [J13], has led to a comprehensive framework published in [”Codes over the Absorption Channel,” IEEE Trans. Inf. Theory, 2024]. This paper simulates neural actions and establishes the theoretical limits and practical coding schemes for communication via a living nervous system. The ultimate objective is to leverage our body’s nervous system for transmitting information between micro-machines within the body.

Future Research

My vision for future work involves a continued commitment to maximizing the storage capacities of existing systems, the development of an efficient DNA-based storage system, a deeper exploration of the dynamics of distributed storage systems, and the formulation of codes that will set the standard for information representation in emerging memory technologies.

DNA-Based Storage:

Building on our latest analysis of k-tuple frequencies, one of my primary goals is to develop coding techniques to correct errors induced by compositions of mutations, moving beyond single-mutation models. We recently achieved progress in this direction when we showed a central limit theorem for the distribution of k-tuples in mutation systems (see the work ["CLT for mutation systems"] which is currently in preparation). This theorem helps us to understand the distribution of k-tuples better, which may be used to construct error-correcting codes. In the in vitro domain, having established the power of constrained coding for nanopore sequencing, I plan to develop practical, low-complexity encoders and decoders that can be integrated into live synthesis pipelines. Furthermore, regarding our work on the optimal reference [J11], I plan to estimate the distribution of the synthesis time, aiming to minimize its variance and ensure a more predictable process.

Distributed Storage Systems:

The exploration of dynamical distributed storage systems is in its nascent stage. I am eager to delve into the study of storage capacity in diverse models, focusing on general models featuring units with distinct failure probabilities and developing coding schemes that leverage the dynamic nature of the system.

Emerging Memory Devices:

My intention is to persist in the exploration of semiconstrained systems and to formulate a comprehensive encoding and decoding scheme for adoption by the industry. Another avenue I am eager to continue is the absorption channel. Building on our 2024 theoretical framework, I aspire to collaborate with scientists to construct a real-life model that communicates via nerves, translating our theoretical coding schemes into a tangible application.

- **Teaching Statement**

  Shaping the minds of the next generation of scientists is a privilege, and I am enthusiastic about the opportunity to teach and mentor students. I view teaching as the theoretical researcher's immediate contribution to society. While theoretical research may take years before practical implementation, young engineers integrating into the tech industry contribute to the progress of the entire country. These engineers serve as the driving force propelling technological development forward, and it is in our best interest that they have a strong and steady educational foundation.

  My appreciation for the power of a good teacher was realized during high school. I attended the Israel Arts and Science Academy from 10th to 12th grade, renowned as one of the best schools in Israel. The diverse attitudes of the teachers and the institute towards education were instrumental to its success. Many teachers, including all the science instructors, were university professors or teaching assistants. Our literature teacher was a renowned writer, and each educator was an expert in their field. This diversity in teaching methods became apparent, prompting me to become a keen observer of my teachers, analyzing and assessing methods to understand what worked best for me. Throughout my undergraduate studies, I continued this analysis, accumulating a collection of effective teaching techniques.

Teaching Philosophy

My teaching philosophy is a fusion of several methods that I have found useful and productive over the years. I have distilled core ideas from each method to construct my teaching style, emphasizing two key principles: preparation and adjustment. Being well-prepared is crucial. Preparation begins before the course starts, with well-organized material and topics arranged in a predefined order. Lecture notes should be comprehensive, and home assignments should include carefully chosen questions contributing to student understanding. A well-prepared teacher significantly enhances student understanding and effort.

However, it's essential to recognize that every class is unique. This brings me to the second key point – adjustment. Recognizing the distinct mix of students in each class, I strive to adapt my techniques to the class rather than the class to my techniques. For example, if there is a majority of math majors, I provide more theoretical and math-oriented examples. Adjusting to the class composition enhances comfort and openness to new information. Additionally, I encourage students to express their thoughts and concerns, making changes as needed to improve the clarity of new material.

In addition to these key principles, I follow a four-step technique when teaching new topics:

1. **Humanization:** Share historical anecdotes related to the discipline to provide context and motivation.
2. **Accuracy and Clarity:** Present precise definitions clearly, focusing on the main point without unnecessary complications.
3. **Intuition:** Explain the intuition behind techniques and theories, emphasizing fundamental concepts with illustrative examples.
4. **Application:** Connect the learned material to real-world applications, demonstrating practical use.

My goal is to equip students with the foundational knowledge, understanding, and intuition necessary for independent exploration. By the end of the course, I aim for students to be selfreliant, possessing a basic set of tools to continue enriching their knowledge with minimal assistance.

Teaching Experience

My pedagogical experience spans over a decade, beginning with mentoring several math students while still an undergraduate. During my graduate studies, I served as a lab instructor for two semesters and as a TA for eleven semesters. Additionally, I held a teaching position at the IDF Flight Academy for three summer semesters, where I fulfilled various roles such as giving recitations, preparing course materials, constructing and grading exercises and exams, and more.

I was also offered a lecturer position in a new university project under the Department of Mathematical Education—a special after-school program for gifted children aged 8-18. Teaching in this program was both challenging and rewarding. I instructed courses that demanded strong abstract thinking, such as logic, topology, finite fields, and more. Teaching these topics to young kids, particularly those aged 8-12, required significant effort to convey abstract ideas in an interesting and engaging manner. This experience provided me with valuable insights and a toolkit that continues to serve me well.

During the Fall semester of 2019, I served as a co-instructor for a graduate-level course on modern discrete probability at the University of Maryland, College Park. As part of my responsibilities, I prepared and delivered some of the course's lectures and assisted in constructing and grading homework exercises and exams. This experience allowed me to practice and enhance my teaching skills in a different language.

In the last years, I have taught several courses at the School of Electrical Engineering. These include a large mandatory undergraduate course (Introduction to Digital Systems), a medium-sized core course (Introduction to Control Theory), and two graduate-level courses (Modern Discrete Probability and Coding for Constrained Systems). Throughout the years, I have received the Glatt award, Dean award, and School award for excellence in teaching.

Teaching Interests

I am excited about the opportunity to construct and teach courses according to the methods and techniques I have mentioned. My teaching interests span all coding/information theory and communication courses as well as courses with a large volume of probability. A partial list follows.

Basic level classes: Discrete math, Intro to probability, Digital systems, Intro. to computer architecture, Intro. to Boolean algebra.

Intermediate classes: Intro. to error correction codes, Coding for constrained systems, Combinatorics of words, Information theory, Stochastic process, Percolation, Random graphs.

Advanced classes: Advanced topics in coding theory, Ergodic theory, Symbolic dynamics, Modern Discrete probability, Stochastic approximation, Large deviations theory, and concentration inequalities.